

Documentation data input interfaz_remix

This document contains the documentation for the chemical data input in the code **interfaz_remix**. There are up to 5 input files described here, ending in “_sist_quim.dat”, “_quim_loc.dat”, “_tar_wat.dat”, “_tar_sol.dat” and “_tar_gas.dat”. The files “_tar_sol.dat” and “_tar_gas.dat” are optional, depending on whether the chemical system contains solids and/or gases. All of the files must contain the **same root** in their names, and they must live in the **same directory**.

We describe the input files with an example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory “..\examples\calcite_eq”.

Chemical system file: “_sist_quim.dat”

This file contains the information related to the chemical system (species, reactions, phases). It is read by the **Chemical System** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name
 - If Section name = ‘PRIMARY AQUEOUS SPECIES’ or ‘AQUEOUS COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
 - If Section name = ‘MINERALS’ or ‘GASES’:
 - i. **Line 3a:** *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - iii. *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘REDOX REACTIONS’:
 - i. **Line 3a:** *react_name* (string), *flag_eq* (logical)
 - i. *react_name*: Name of redox reaction in database “*Monod_DB.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘SURFACE COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
- **Line 4:** Section separation
- **Line 5:** end of file

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'PRIMARY AQUEOUS SPECIES'

'h2o(p)' ! name of primary aqueous species

'h+'

'ca+2'

'o2(aq)'

'hco3-'

'*'

'MINERALS'

'calcite' .true. .true. ! name of mineral, flag equilibrium, flag constant activity

'*'.true. .true.

'GASES'

'o2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'co2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'*'.true. .true.

'-----'

'end'

Local chemistry file: “_quim_loc.dat”

This file contains the **local** chemical information (water types, mineral zones, gas zones, surfaces). It is read by the **Chemistry** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Section name:** ‘INITIAL AND BOUNDARY WATER TYPES’
 - **Line 1:** *act_coeffs_model* (integer)
 - 0. Ideal
 - 1. Debye-Hückel
 - 2. Debye-Hückel extended
 - 3. Davies
 - 4. Truesdell-Jones
 - **Line 2:** *num_wat_types* (integer)
 - i. Number of water types (includes initial, recharge, boundary, etc)
 - **Line 3:** *ind_wat_type* (integer), *temp* (real)
 - i. *ind_wat_type*: index of water type
 - ii. *temp*: temperature of solution (in Celsius)
 - **Line 3a:** *name* (string)
 - i. Name of water type
 - **Line 3b:** water type data begins
 - i. *icon*: initial condition type
 - 1. prescribed primary concentration
 - 2. prescribed total concentration
 - 3. prescribed activity
 - 4. primary species is in equilibrium with a constrain (mineral, gas or aqueous complex)
 - ii. *guess*: Initial guess for primary species concentration
 - iii. *ctot*: (all concentrations are expressed in molalities)
 - i. *icon*=1: primary species concentration
 - ii. *icon*=2: total concentration of primary species
 - iii. *icon*=3: activity of primary species
 - iv. *icon*=4: activity of constrain that is in equilibrium with this primary species
 - iv. *constrain*: if *icon*=4, it is the species in equilibrium with the primary species
 - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
 - **Line 3d:** end of water type data
 - **Line 4:** end of section

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'INITIAL AND BOUNDARY WATER TYPES'

0 ! activity coefficients model

2 ! number of water types

1 25.0 ! index water type, temperature (C)

'boundary' ! name of water type

' icon guess ctot constrain'

'h2o(p)' 3 1d0 1d0 "

'ca+2' 1 1d-9 1d-9 "

'o2(aq)' 4 1d-9 2d-1 'o2(g)'

'hco3-' 4 1d-9 3.5d-4 'co2(g)'

'h+' 1 1d-7 1d-7 "

'*' 0 0.0 0.0 "

2 25.0 ! index water type, temperature (C)

'resident' ! name of water type

' icon guess ctot constrain'

'h2o(p)' 3 1d0 1d0 "

'ca+2' 1 1d-3 1d-3 "

'o2(aq)' 1 1d-6 1d-6 "

'hco3-' 4 1d-9 1d0 'calcite'

'h+' 1 1d-5 1d-5 "

'*' 0 0.0 0.0 "

'-----'

- Section name: 'INITIAL MINERAL ZONES'
 - **Line 1:** *num_min_zones* (integer)
 - i. Number of mineral zones
 - **Line 2a:** *ind_min_zone* (integer), *temp* (real)
 - i. *ind_min_zone*: index of mineral zone
 - ii. *temp*: temperature of mineral zone (in Celsius)
 - **Line 2b:** *name* (string)
 - i. Name of mineral zone
 - **Line 2c:** mineral zone data begins
 - **Line 2d:** *min_name* (string), *vol_frac* (real), *area* (real)
 - i. *min_name*: name of mineral in mineral zone (must be in the chemical system)
 - ii. *vol_frac*: initial volumetric fraction of mineral
 - iii. *area*: specific surface of mineral (surface of mineral per unit volume of rock)
 - **Line 2e:** end of mineral zone data
 - **Line 3:** end of section

'INITIAL MINERAL ZONES'

1 ! number of mineral zones

1 25.0 ! index of mineral zone, temperature (C)

'calcite' ! name of mineral zone

'mineral vol.frac. area'

'calcite' 0.50 1d2

'*' 0.00 0.00

'-----'

- Section name: 'INITIAL AND BOUNDARY GAS ZONES'
 - **Line 1:** *num_gas_zones* (integer)
 - i. Number of gas zones
 - **Line 2a:** *ind_gas_zone* (integer), *temp* (real), *vol* (real)
 - i. *ind_gas_zone*: index of gas zone
 - ii. *temp*: temperature of gas zone (in Celsius)
 - iii. *vol*: total volume of gas
 - **Line 2b:** gas zone data begins
 - **Line 2c:** *name* (string)
 - i. Name of gas zone
 - **Line 2d:** *gas_name* (string), *part_press* (real)
 - i. *gas_name*: name of gas in gas zone (must be in the chemical system)
 - ii. *part_press*: initial partial pressure (equivalent to activity) of gas (in atm)
 - **Line 2e:** end of gas zone data
 - **Line 3:** end of section
 - **Line 4:** end of file

'INITIAL AND BOUNDARY GAS ZONES'

1 ! number of gas zones

1 25 1d9 ! index gas zone, temperature (C), volume

'atmosphere' ! name of gas zone

'gas partial pressure'

'o2(g)' 2d-1

'co2(g)' 3.5d-4

'*' 0.00

'-----'

'end'

Waters file: “_tar_wat.dat”

This file contains the **association** between waters and water types on one hand, and between target waters, target solids and/or target gases on the other hand. It also indicates if the waters are external (i.e. recharge or boundary) or not. This file is read by the **Chemistry** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name: 'WATERS'
 - **Line 3a:** *num_wat* (integer)
 - Number of waters
 - **Line 3b:** *num_rech_wat* (integer)
 - Number of recharge waters
 - **Line 3c:** *num_bd_wat* (integer)
 - Number of boundary waters
 - **Line 3d (repeated as many times as intervals are present):** *int_wat* (string), *iwtype* (integer), *int_sol* (string), *int_gas* (string), *flag_wat_type* (integer)
 - *int_wat*: Interval of waters
 - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
 - *int_sol*: Interval of target solids associated with *int_wat* (0 means no associated target solids)
 - *int_gas*: Interval of target gases associated with *int_wat* (0 means no associated target gases)
 - *flag_wat_type*: Flag to indicate water type.
 - 0: Boundary
 - 1: Domain
 - 2: Recharge
 - End of section
- **Line 4:** end of file

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'WATERS'

12 ! number of waters

0 ! number of recharge waters

2 ! number of boundary waters

1 1 0 1 0 ! waters interval, water type index, target solids interval, target gases interval, flag water type

2-11 2 1-10 0 1

12 2 0 0 0

'end'

Target solids file: “_tar_sol.dat”

This file contains the **association** between target solids and solid zones. Each target solid corresponds to a target (i.e. cell/node/element) in the mesh, therefore the number of target solids cannot be larger than the number of targets. This file is read by the **Chemistry** class. The only solid zones implemented so far are the **mineral** and **exchange** zones. **IMPORTANT:** this file will not be read by the code if there are no solid zones.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name: ‘TARGET SOLIDS’
 - **Line 3a:** *num_tar_sol* (integer)
 - Number of target solids
 - **Line 3b:** *interval* (string), *iszone* (integer)
 - *interval*: Target solids interval
 - *iszone*: Index of mineral zone in section ‘INITIAL MINERAL ZONES’
 - end of section
- **Line 4:** end of file

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'TITLE OF THE PROBLEM: Calcite in equilibrium'
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'-----'
```

```
'TARGET SOLIDS'
```

```
10                                ! number of target solids
```

```
1-10    1                        ! target solids interval, solid zone index
```

```
'-----'
```

```
'end'
```

Target gases file: “_tar_gas.dat”

This file contains the **association** between target gases and gas zones. Each target gas corresponds to a target (i.e. cell/node/element) in the mesh, therefore the number of target gases cannot be larger than the number of targets. This file is read by the **Chemistry** class. **IMPORTANT:** this file will not be read by the code if there are no gas zones.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name: ‘TARGET GASES’
 - **Line 3a:** *num_tar_gas* (integer)
 - Number of target gases
 - **Line 3b:** *interval* (string), *igzone* (integer)
 - *interval*: Target gases interval
 - *igzone*: Index of gas zone in section ‘INITIAL AND BOUNDARY GAS ZONES’
 - end of section
- **Line 4:** End of file

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'TARGET GASES'

1 ! number of target gases

1 1 ! target gases interval, gas zone index

'-----'

'end'